

PACT: An Accountability Protocol for Deployed AI Systems, with Measured Limits on AI-Jury Adjudication

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Abstract—Institutional oversight of AI rests on a weak foundation. Third-party evaluation, statutory record-keeping and proposals for verifiable limits on frontier development all presume an audit record. That record is produced and held by the operator of the system under audit. We present PACT, a protocol that makes deployed AI conduct non-repudiable and adjudicable among mutually distrusting parties. PACT separates a cheap evidence plane, consisting of per-agent hash-chained conduct logs whose Merkle checkpoints are cosigned by independent witnesses, from a small governance plane holding registries, challenges, verdicts and consequences. Conduct is policed recursively: watchdog agents file challenges, diversity-constrained juries of evaluator models vote by commit–reveal and humans hold final appeal. Jurors are themselves covered by the mechanism. We built a reference implementation that runs the full loop at 122 μ s per covered action and costs 13.3 B of shared ledger per query. In it, one action in 100,000 remains provable to a witness-cosigned root through 22 hashes. Our central results are negative and were produced by that implementation against its own design. Auditing our own corpus against its policy clause invalidated 11 of its 32 violation labels, so every number below is reported on the corpus as measured and on the audited corpus, the audited figure primary. A two-thirds jury scores below its best single member on both (gap -0.321 , 95% CI $[-0.507, -0.144]$ as measured; -0.117 $[-0.289, 0.000]$ audited, never strictly better in 20,000 replicates), because majority voting outvotes a lone correct detector. An asymmetric rule holds containment of true violations at 0.905 $[0.762, 1.000]$ where the symmetric rule leaks 71.4% of them, at the price of 71.2% human review load. Errors of three same-vendor evaluator models are correlated at $4.7\times$ what independence predicts as measured (95% CI $[3.2, 9.5]$) and $5.8\times$ $[3.7, 12.8]$ audited, every pairwise test rejecting independence, concentrated on the cases we pre-labelled as contested. A nineteen-juror replication across nine vendors then removes the explanation we had offered for that number: cross-vendor pairs are as correlated as within-vendor pairs (difference 95% CI contains zero on both corpora), so the correlation is a property of the items and of model judgment rather than of shared lineage, and every aggregation rule scores below the best single juror — findings that replicate under a second prompt framing. This agrees with concurrent findings on generic judge panels [39], [40]; we measure it where a verdict carries consequence and we change the protocol in response. The diversity constraint on which AI-jury designs depend buys less than it claims and we restate it as selection for measured complementarity.

Index Terms—AI accountability, transparency logs, LLM-as-judge, distributed systems security, adjudication protocols, empirical evaluation

I. INTRODUCTION

Oversight of AI systems is acquiring an institutional layer faster than a technical one. Regulators impose record-keeping and incident-reporting duties on high-risk systems [18], developers publish safety frameworks and invite external evaluation. Policy proposals call for verification arrangements spanning parties that do not trust one another. Every such mechanism presumes a trustworthy account of what an AI system did. In current practice that account is produced, stored and served by the operator of the system under audit, so the record is editable by the party it constrains.

Transparency logs solved the analogous problem for web certificates [1], but detection-only machinery is insufficient here. AI oversight also needs a place to challenge recorded conduct, a procedure to adjudicate the challenge and a mechanism that binds consequences to the verdict. These are stateful protocol functions that a passive log cannot provide.

A recurring proposal is to have AI systems adjudicate each other, using panels of evaluator models drawn to be diverse and voting on whether conduct violated policy [10], [12], [38]. Human review does not scale to the volume of AI actions, so the appeal is clear. The assumption underneath is that jurors drawn from different model families fail independently, so that aggregating them buys reliability. Recent work has begun to question that assumption on generic evaluation benchmarks [39], [40]. What has not been examined is how it behaves inside a protocol that attaches consequences to the verdict, where the design must then respond. We build the protocol such a design needs, we measure the assumption within it and we change the protocol in light of the answer.

Contributions.

- **C1: Protocol (§III).** We define six accountability properties for deployed AI systems and give a two-plane design (Fig. 1) that achieves them among mutually distrusting parties: hash-chained conduct logs with witness-cosigned checkpoints and a minimal governance plane for registries, challenges, verdicts and consequences.
- **C2: Adjudication at scale (§IV–V).** We combine diversity-constrained jury sampling with commit–reveal voting and first-class abstention, a precedent graph that amortizes repeated adjudications and risk-limiting audits

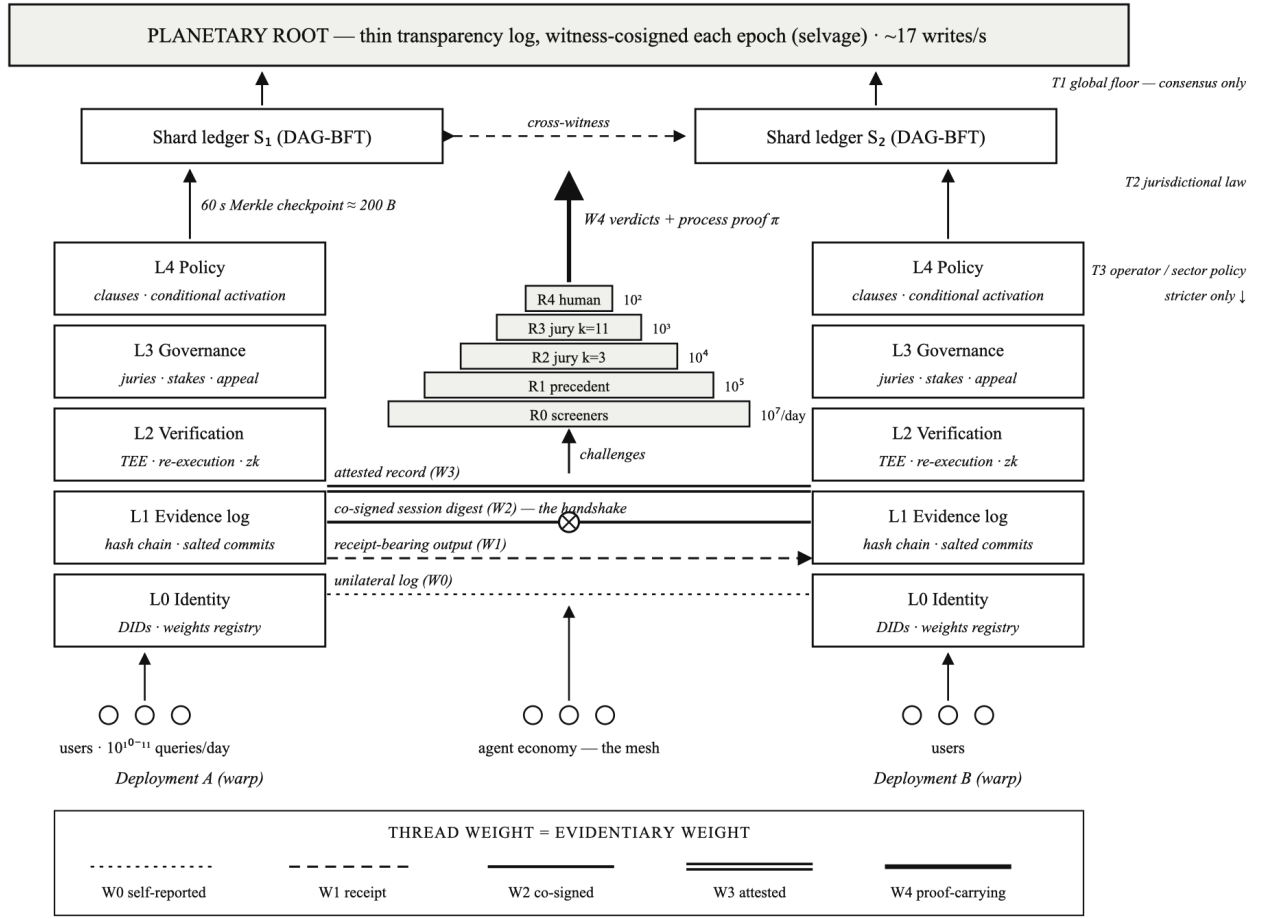


Fig. 1. The accountability loom. Vertical warp columns are per-deployment stacks; horizontal weft threads are interaction evidence, drawn with thread weight proportional to evidentiary class; the central funnel is the shuttle turning flagged threads into verdicts; the witness-cosigned root is the selvage. The right-hand rail marks the three policy tiers at their native altitude.

that bound violation rates from constant-size samples independently of screener error.

- **C3: Federation without global consensus (§IV).** Per-agent logs commute, so evidence needs no total order. Agent, operator, shard and witness-cosigned root compose into inclusion proofs that keep every individual action provable while ledger cost stays flat in traffic.
- **C4: An evaluation that refutes its own design (§VI–VII).** The claims we began with keep failing measurement: jury precision, juror error-independence, eleven of our own corpus labels, and finally the shared-lineage explanation we offered for the correlation, which a nineteen-juror, nine-vendor replication removes. Each failure is reported beside the protocol change it forced.

II. THREAT MODEL AND ACCOUNTABILITY PROPERTIES

Principals. Model developer D , deployer/operator P , auditor A , watchdog agent W , juror model J , shard validator V , log witness T , human review board H .

Adversaries. We consider a cheating operator who serves unregistered weights, omits or tampers with entries or equivocates between audiences; watchdogs who spam frivolous challenges; jury subsets who collude; validators who censor; and observers who mine commitments for private content.

Assumptions. At most f Byzantine validators among $3f+1$ per shard [7]. At least one honest witness per cosigning quorum. Hardware attestation is trusted to vendor level, which is an explicit trust anchor rather than a guarantee. Re-execution nondeterminism is bounded and measurable.

Non-goals. PACT cannot conscript systems served outside instrumented runtimes. It proves the integrity of covered conduct, never the absence of uncovered conduct. It does not verify training-time claims, for which compute accounting [4] is the complement.

We define a *covered action* as an inference or tool call served through a PACT-instrumented runtime. For every covered action PACT provides: P1 Attribution to (model version, weights commitment, runtime measurement, operator); P2 Integrity, since tampering breaks a hash chain against cosigned

checkpoints; P3 Non-repudiation, since entries are signed by attested runtime keys; P4 Challengeability within window Δ against a policy clause; P5 Adjudicability, since every challenge terminates in a recorded verdict; and P6 Consequence binding, since verdicts mechanically update stake, reputation and registry status, which deployment gates read.

III. THE PROTOCOL

A. Identity and registry

We use decentralized identifiers for all principals. Per model version the registry records a Merkle root over weight shards, a training-data manifest commitment, a lineage link, a model-card hash [13], the policy hash served under and approved runtime measurements. Agent-instance keys are generated inside attested enclaves, so one signature transitively binds output to runtime image to registered weights to operator.

B. Evidence plane

Each agent maintains a hash-chained log. An entry carries a sequence number, a previous-hash link, salted commitments to input and output, tool calls, the policy clause, the sampling seed and a signature. Per-entry salts are operator-held and escrow-shared with in-remit auditors. They defeat dictionary attacks on hashed personal data. Deleting payload and salt makes the on-ledger commitment permanently unlinkable, which is how we support erasure. Every T seconds or N entries we checkpoint the log’s Merkle root, independent witnesses cosign the tree head [21] and inclusion and consistency proofs follow transparency-log practice [1].

Two implementation details are load-bearing and we got both wrong in our first version (§VII-E). Odd nodes are promoted rather than duplicated, so leaf count is unambiguous from the root. Inclusion proofs carry sibling hashes only: the side of each sibling and the required path length are re-derived at verification from the claimed index and tree size. A proof therefore cannot be replayed at a position other than the one it was issued for.

C. Verification ladder

Verification climbs three rungs, ordered by cost. Attestation binds the log to registered weights and an approved runtime image. We enforce registry-approved measurements at verification and an agent with no registered runtime is reported unattested rather than passing silently. Optimistic re-execution re-runs challenged inferences from logged seeds, with a fraud-proof-style bisection [6] narrowing disputes to single entries. Bitwise cross-hardware reproducibility is unattainable, so acceptance is within a logit tolerance decided by quorum. Zero-knowledge inference proofs [5] remain confined to small judge models by cost. We implement only the first rung here; the other two are specified and unmeasured.

D. Governance plane and the jury

Watchdogs file challenges naming a log range, a policy clause, evidence commitments and a bond. A verifiable random function samples a jury of $k = 11$ evaluator models under

diversity constraints: at most one juror per operator and at least three base-model families. We take the seed from the root of the first checkpoint sealed after the challenge was filed. That value did not exist when the challenge id was assigned, so a challenger cannot grind for a favourable panel.

Voting is commit–reveal. Each juror publishes $H(\text{verdict} \parallel H(\text{rationale}) \parallel \text{nonce})$, where the nonce is 128 bits of juror-held randomness. A deterministic nonce leaves the hiding property absent while the ceremony still looks correct (§VII-E). A two-thirds supermajority decides. Splits and severity $\geq S2$ escalate to the human board.

Measurement rather than design taste forced two refinements. The first is the lone-detector rule. Plain majority voting is symmetric and symmetry is wrong for safety adjudication, because when a harm is visible to only one juror family the majority can vote down the juror that was right. For catastrophic clauses we therefore allow a single violation vote to be escalated but never overridden into an auto-clear. The second is that abstention is a verdict rather than a fault. A juror may decline a case, its silence counts toward neither side and we never slash it. A panel thinned below quorum or below its family floor escalates instead of deciding on the remainder. We punish only equivocation, meaning committing one verdict and revealing another, abstention included.

Severity gates on evidentiary weight, so a conviction at $S1$ or above may not rest on unilateral self-reported evidence. Slashing is asymmetric: operator stake on proven violation, challenger bond on frivolity and juror stake only for provable process faults. We never slash a juror for voting with the minority, which would breed herding. Verdicts carry the jury evidence that produced them and verification re-derives the draw, so a verdict written directly to the store is reported as unbacked rather than silently trusted. Enforcement teeth come from gates: serving platforms, marketplaces and insurers read registry status, mirroring the way browser enforcement gave certificate transparency its force. The design is recursive, since jurors and watchdogs are covered models whose verdicts are covered actions.

IV. FEDERATION WITHOUT GLOBAL CONSENSUS

Per-agent logs are independent hash chains. They commute, so evidence needs no total order. Only governance actions require consensus and they are rare. The hierarchy runs from agent log to operator root to shard ledger to planetary root, where the planetary root is a thin transparency log of shard roots cosigned each epoch by an independent witness fabric rather than a blockchain.

Aggregation. Batching moves bandwidth, not trust. Each hop’s root reappears as the next hop’s leaf, bound to its owner’s identity, so a valid subtree cannot be re-claimed under a different identity. Every individual action stays provable to the root by a composed $O(\log n)$ path and the BFT thresholds, witness quorums and salted-commitment privacy are identical at every scale.

Cross-witness gossip. Equivocation dies at the root, but only if witnesses talk to each other. Measurement sharpened

this into a protocol requirement (§VII-D). A deceived witness sees one internally consistent history and has nothing to object to, so equivocation is invisible from any single vantage point and exists only as a disagreement between vantage points. We therefore make cross-witness gossip mandatory. A fabric that cosigns without comparing detects no equivocation whatever its size.

V. ADJUDICATION AT SCALE

We treat full juries as the last resort rather than the first responder. Flagged conduct passes through co-located screeners, then a precedent matcher, then a small screening jury, then the full jury, then humans. Each rung has an early exit and an appeal right upward.

Precedent graph. The graph makes every adjudicated case a citable node carrying evidence commitments, clause, verdict and rationale. Matching is embedding similarity plus clause identity, matches auto-resolve with appeal rights and conflicting precedents force escalation, which is how doctrine evolves. Precedents are themselves covered artifacts, supersedable by full-jury or human rulings, so early errors do not fossilize. Measurement forced us to state two things precisely. Pattern count bounds the reduction, not the mechanism: adjudicating N distinct patterns costs at least N juries, so any large factor is a claim about traffic repetitiveness rather than about PACT. The similarity threshold is also a safety parameter. Set too loose, benign text that merely shares vocabulary with an adjudicated violation inherits its verdict. Set too tight, genuine repeats stop matching and the reduction collapses.

Risk-limiting audits. Imported from election auditing [24], these verify a VRF-drawn, Merkle-proven sample sized to a stated confidence. They yield claims of the form “with 99% confidence this model’s violation rate is below ϵ ”, with sample size independent of traffic volume. Because sampling draws from all covered actions, this is the only check independent of screener error and a screener miss is otherwise unrecoverable by any downstream mechanism.

VI. IMPLEMENTATION

Our reference implementation is open source and runs the full loop: evidence log with witness cosigning, jury protocol, precedent graph, trained screener, federation tiers and a serving shim that logs completions and returns a receipt. We make experiments deterministic by hash-seeding rather than RNG-seeding, so every reported number reproduces exactly. We compute intervals as percentile bootstrap over items with 20,000 replicates (`m0/ci.py`), taking the item as the unit that would change if the corpus were rewritten; each replicate resamples the items once and scores every juror pair on that same resample, since an earlier construction that resampled each pair independently averaged away item-level covariance and understated the mean-ratio interval, a correction we report rather than bury. Independence is tested against the permutation null that fixes each model’s error count and randomises which items it misses; that null is hypergeometric, so we report its exact tail. No interval here covers variation from model

choice, prompt framing or policy clause, since those are single draws.

Our evaluation domain is repository secret handling: 70 one-line code-review items under a policy clause prohibiting credential literals in source, committed key material, unredacted credential logging and credentials transmitted with certificate verification disabled. The domain is benign by construction and the findings below are properties of the voting rule rather than of any domain.

The construction matters more than the domain. We wrote five detector families from the policy clause alone and committed them before any data item existed. Six separate language-model agents then wrote the items from the same clause, each prompted in isolation, without access to the detector code and without sight of each other’s output. Neither side could tune to the other. In earlier versions of this work the same author wrote both, which made the families’ blind spots an artifact of authorship and inflated every result derived from them.

We stress that these authoring agents are instances of related models rather than independent human authors. This matters for §VII-C, where our jurors are also models of a related lineage, so the items we label contested may be contested for that lineage in particular. We treat it as a confound on the magnitude of the correlation we report and return to it in §VIII. Concurrent measurements of the same effect on corpora we had no part in building [39], [40] are why we read the direction of the result as sound despite it.

Construction discipline did not save the labels themselves. Our safety rule wrote every credential value as an announced placeholder (`ghp_SYNTHETIC_NOT_REAL`), while the clause prohibits a *live* credential written as a literal — and a value that announces itself synthetic is not live. Eleven of the 32 violation labels rest on exactly that bullet and conflict with the clause as written; a model that clears them is following its instructions. The audit that found this (`m0/label_audit.py`) is a function of item text and clause only, computed blind of every verdict so it cannot be tuned to excuse whichever items models missed, and it is conservative: the committed-key items keep their labels, because that bullet carries no live qualifier. We exclude the eleven items rather than relabel them, pending adjudication of the labels themselves, and report every label-dependent number twice — on the 70 items as measured and on the 59-item audited corpus (21 violations) — with the audited figure primary where they disagree.

VII. EVALUATION

Our evaluation runs in one process, with development keys held beside the data rather than in an enclave, simulated rather than hardware attestation, federation tiers as objects rather than hosts and a synthetic corpus. These numbers measure the protocol, not deployment realities.

A. Append path and footprint

A naive implementation costs 567 μ s p50 per covered action. Three changes bring it to 122 μ s p50 and 142 μ s

p95 (7,800 entries/s single-threaded in Python): write-ahead logging 2.2 $\times$, caching the deserialized signing key 1.7 $\times$ and batched commits 1.2 $\times$, for 4.6 $\times$ end to end. We made each independently toggleable, so we measure the attribution rather than assert it. Profiling shows that 76% of the remainder is Ed25519 signing, which is the non-repudiation guarantee itself. The path is therefore signature-bound and further gains require batch-signing rather than micro-optimization.

An entry is about 406 B in the operator’s own log and a witness-cosigned checkpoint about 663 B. The shared cost is 13.3 B per query at an aggressive 50-entry checkpoint interval, falling to about 0.07 B per query at 10⁴-entry batching. Ledger cost is a function of time, not of traffic.

B. Adjudication quality and the central negative result

Single detectors span F1 0.111 to 0.577 as measured; audited, the spread is 0.160 to 0.414 and the best family changes from semantic to contrastive, because nine of the eleven conflicted items were semantic’s true positives. The audit strips the easy positives — the placeholder markers sat on exactly the shape-matched literals these families key on — so rule-based detection gets harder on the clean corpus, not easier. Our scoring makes no assumption about human accuracy: we report the machine’s three-state output rather than crediting escalations with ground truth. An earlier version of this harness did credit them, which inflated a headline into a number that mostly measured that assumption.

Under a plain two-thirds rule the machine auto-clears 50.0% of true violations as measured and 71.4% audited (containment 0.500, 95% CI [0.312, 0.656]; 0.286 [0.095, 0.476]), convicting at precision 0.714 and 0.667 while sending 20.0% and 8.5% of traffic to a person. The lone-detector rule holds containment at 0.906 [0.781, 1.000] and 0.905 [0.762, 1.000], at human load 72.9% and 71.2%. On the audited corpus the two rules’ containment intervals are disjoint: correcting the labels makes the case for the asymmetric rule stronger, because the symmetric rule’s apparent containment was riding on violations that were easy to convict only because they were mislabelled. Machine-level precision and recall are identical under both rules. The rule changes nothing about what the system decides, only about what it refuses to clear silently, so we report containment rather than F1 for it.

Scored as machine decisions, the symmetric jury reaches F1 0.256 against its best single member’s 0.577 as measured, a gap of -0.321 with bootstrap 95% CI $[-0.507, -0.144]$. Audited, the gap is -0.117 $[-0.289, 0.000]$: the jury is never strictly better in any of 20,000 replicates and exactly ties in about 13%, but the interval touches zero, so at this corpus size the audited rule-based data alone cannot carry the claim. The ties are themselves the finding — the audited jury’s decisions coincide with the single structural detector on all 59 items, and on 69 of 70 as measured, so the eleven-seat ensemble behaves as one member. We keep the conclusion that majority voting is not automatically safer than its best member, and rest its statistical weight on the real-model panels of §VII-C, where every aggregation rule scores below the best single juror.

One claim from an earlier draft does not survive. We previously held that a diverse jury raises precision over its best member. Here precision falls, 0.750 to 0.714 as measured and 0.750 to 0.667 audited. The original gain existed because that corpus concentrated its false positives in two of five families, where majority voting washed them out. And the complementarity these families do have is thinner than we first printed: five of ten pairs miss violations the other catches as measured, four of ten audited, with two miss-sets becoming identical — an earlier count that included one-directional overlaps overstated it. Aggregation failed to improve precision anywhere. Complementary coverage is necessary for a jury to pay off and is not sufficient.

C. Do evaluator models fail independently?

The diversity constraint is worth its cost only if jurors’ errors are uncorrelated. Rule-based detectors cannot test this, so we put the same 70 items to three real model tiers, blind, under one policy clause. All three responded and all three made real errors: 21, 8 and 6 of 70 as measured, and 10, 7 and 5 of 59 audited. The eleven conflicted items account for eleven of the weakest tier’s errors but only one each of the other two’s — under that framing the stronger tiers sided with the label and the weakest with the clause, which is one more reason we exclude those items rather than score anyone against a label in dispute.

We find errors correlated at 4.7 $\times$ what independence predicts as measured (95% CI [3.2, 9.5]) and 5.8 $\times$ [3.7, 12.8] audited, above 1.0 in every bootstrap replicate on both corpora. Correcting the labels strengthens the correlation — the opposite of what a labelling artifact would predict, because the items removed were ones jurors got “wrong” together only by following the clause. The pairwise ratios are 2.5 $\times$ [1.4, 4.0], 2.8 $\times$ [1.6, 4.4] and 8.8 $\times$ [5.0, 23.3] as measured; 4.2 $\times$ [2.2, 8.4], 4.7 $\times$ [2.3, 9.8] and 8.4 $\times$ [4.9, 19.7] audited, the last pair in each the two closest-lineage models and the least precisely estimated. The permutation null that fixes each model’s error count and randomises which items it misses has an exact hypergeometric tail, and it rejects independence for every pair on both corpora (largest exact $p = 0.0080$ as measured and 0.0021 audited; the closest-lineage pair is below 0.00001 on both). Five items defeat all three models as measured and four audited (7.1% of items, 95% CI [1.4%, 14.3%]; 6.8% [1.7%, 13.6%]), and on both corpora every one is from the pre-labelled contested set — though at four or five items that claim carries a wide interval and is the weakest in this section. The models converge on the easy calls and make the same error on the hard ones. Agreement is cheap where the models agree and their errors correlate where the verdict matters, so a supermajority of correlated jurors is confidently wrong rather than usefully uncertain.

In an earlier draft we read that number as an upper bound: same-vendor models share tuning lineage, so lineage might be inflating the correlation, and we called a cross-vendor replication the obvious next experiment. We then ran it, and the reading does not survive. Nineteen jurors from nine ven-

dors — OpenAI, Google, Meta, Mistral, Alibaba, DeepSeek, Anthropic, xAI and Amazon — judged the same items blind under one fixed prompt whose hash is stamped into the verdict file (`m0/juror_prompt.py`, `m0/run_jurors.py`), giving twelve within-vendor pairs against 159 cross-vendor pairs (`m0/e15.py`). Cross-vendor pairs are as correlated as within-vendor pairs: mean $3.2\times$ against $3.1\times$ as measured and $4.1\times$ against $4.0\times$ audited, the within-minus-cross difference $-0.1\times$ with a 95% CI containing zero on both corpora ($[-0.3, 0.1]$; $[-1.3, 1.3]$). All 171 pairs reject independence as measured; audited, 102 of 171 do at $p < 0.05$ — each juror makes only three to twelve errors there, so the per-pair test loses power even as the mean ratio rises — and the bootstrap puts the cross-vendor mean above 1.0 in every replicate on both corpora. Shared lineage is not what produces the agreement. We withdraw the upper-bound reading: the correlation is a property of the items and of how models read them, and on the audited corpus no single item defeats the full nineteen-model panel, so the agreement is pairwise and shifting rather than a fixed set of impossible items.

The nineteen-juror panel also re-asks §VII-B’s question with real jurors, and the answer is sharper. Every aggregation rule scores below the best single juror on the audited corpus: two-thirds 0.778, simple majority 0.842, any-single-vote conviction 0.875, against 0.927 for the best juror alone. As measured, the any-single-vote rule appears to beat the best juror, 0.882 against 0.800 — an artifact of the conflicted labels, which reward whichever rule fires most often, and a caution we now read back onto every aggregation result computed on unaudited data. The asymmetric rule’s value is containment, which survived every cut we tried; its value is not accuracy, which survived none.

Because prompt framing is the one variable our intervals cannot cover, we reran the full nineteen-juror panel under a second, deliberately minimal framing (`jp:v2`, hash-stamped like the first). Both findings replicate: the within-minus-cross difference again contains zero on both corpora ($-0.0\times [-0.2, 0.2]$; $-0.2\times [-1.5, 1.3]$) and, audited, every aggregation rule again scores below the best single juror (0.811, 0.900 and 0.824 against 0.977). What the framing does move is nearly every individual number — mean correlation, per-juror quality, even which rule looks best on the unaudited corpus — so we report the two invariants as the findings and treat any single panel’s point values as framing-specific.

We can state a sharper rule than the one we started with. The pairwise ratios differ substantially, so the correct constraint selects jurors for measured complementarity rather than declared family membership — and vendor membership, we now know, was never the operative variable. Our scope is one corpus, one clause and verdicts at temperature zero, with prompt framing mattering more than we first allowed: the three-tier run predates this paper’s prompt discipline, its exact framing was not recorded, so those specific numbers are not strictly reproducible, and under the framing we now fix and hash, tiers of the same vendor score differently. Every panel since is stamped with the prompt hash that produced it. Concurrent

work reaches the correlation conclusion by different routes and at larger scale, across 350+ models [39] and on a nine-judge panel spanning seven families where the best single judge matches or beats the whole panel [40]. What is ours is the setting and the comparison: verdicts that move stake and registry status, and within- against cross-vendor pairs on identical items.

Juror availability. Availability falls as severity rises. In a companion run on a more sensitive corpus, two of three models refused the adjudication task outright, reporting safeguard categories. A juror must read what it judges, so models tuned to refuse engagement with dangerous content also refuse to moderate it. The panel thins toward the least cautious models exactly where care is most needed and a three-family requirement can become unsatisfiable at top severity. This is why we make abstention first-class in §III-D. Any deployment claim must report severity-stratified abstention rates, because a jury that silently thins at high severity still returns a verdict.

D. Federation, precedent and mesh coverage

Federation. Root cost per epoch is a function of shard count alone, at about 9.7 KB for 10 shards with 10 witnesses, so no query rate moves it. Sealing an epoch over 10,000 operators and 100,000 agents costs 0.48 s, which is 0.8% of a 60-second pin budget, because aggregation hashes roots rather than payloads. One action out of 100,000 remains provable through 22 hashes in a 2.3 KB proof verified in 36 μ s, growing logarithmically. A valid subtree re-claimed under another identity is rejected. Witness quorum degrades linearly under churn and the log records the shortfall rather than failing silently.

Equivocation. A shard presenting a forged root to 3 of 10 witnesses was detected 0 times by any witness individually and 21 times by witnesses comparing, which is exactly the 3×7 deceived/honest pairs. The zero is the result. Locally consistent deception is locally undetectable, which is why we make gossip mandatory.

Precedent. Replaying 174 held-out actions over 29 distinct patterns, the precedent graph reduces jury convocations from 48 to 8 and human reviews from 37 to 7. Patterns that reach the funnel set the binding ceiling and 8 of 29 survive the screener, so the achievable reduction is $6.0\times$. The mechanism meets it exactly. The threshold behaves as a safety parameter: over all 174 actions the closest same-bucket pair sits at cosine 0.998 and the closest opposite-label pair at 0.916, a narrow band, because a private key in a test fixture and the same key in production configuration are textually near-identical and oppositely labelled. At thresholds ≤ 0.55 precedent commits 6 wrong auto-verdicts and at ≥ 0.65 it commits none.

Screener. This is the weak point, with a caveat about how firmly. A trained classifier recalls 0.500 on unseen phrasings in the headline split and whatever it misses is auto-cleared and never reaches adjudication. Across eight equally valid splits, however, recall ranges from 0.167 to 0.500, so the direction of the finding is solid and its magnitude is not. Either way

this is the strongest argument in our results for risk-limiting sampling, which is independent of screener error.

Mesh coverage. Coverage does not percolate. On 2,000-agent random and scale-free interaction graphs, a reciprocal witnessing mesh gives coverage that is one-hop and degree-weighted rather than viral. The coverage curve shows no threshold and in four-hop workflows “all hops observed” trails “at least one hop” badly, at 0.045 versus 0.338 at $p = 0.10$, so workflows are seen in fragments. Exposure follows $1 - (1 - p)^{\text{degree}}$: at $p = 0.10$, degree-16+ agents are exposed 86% of the time against 17% for degree-1–2. Instrumenting the top 1% of agents by degree covers 22% of interactions, the reach random instrumentation needs about 12% to achieve.

E. Integrity of the artifact itself

Because we wrote the code that produced these results, we subjected the paper and implementation to a structured adversarial review. Seven independent reviewers worked with distinct lenses, each required to execute the code and a skeptic tasked with refuting the findings followed each one. Thirty findings survived refutation.

The value concentrated in a class we could not have found by self-inspection, namely properties the paper asserted that the implementation did not enforce. Merkle proofs accepted an index the verifier never read, so a valid proof verified at any claimed position. A session-record verifier stored no signatures and re-signed at verification time, so fabricated records verified successfully. Commit–reveal derived its nonce from public inputs over a two-element verdict space, so every vote was recoverable before any reveal. We fixed all three, each with a regression test verified to fail when the fix is reverted. That check also found three pre-existing tests which could not fail at all, including one that would have permitted the supermajority threshold to be relaxed to a simple majority. Our suite is 49 checks with 23 such mutation-verified detections.

The review was equally useful against our own reporting. It established that the corpus underlying the detection results had been written from the detectors’ own word lists, that a flagship end-to-end figure assumed perfect human review, that a reduction ceiling used a denominator counting patterns no jury ever sees and that a quoted similarity landmark was hand-picked rather than searched. We corrected each above and each correction moved a number downward. We report this history because presenting the corrected numbers as though they were the first ones would misrepresent how much of this evidence depended on our own choices about what to measure.

VIII. LIMITATIONS

Coverage is the central limit, since covered systems become accountable and uncovered ones do not. The attestation anchor is the hardware vendor. Rungs two and three of the verification ladder are specified and unimplemented, so the nondeterminism tolerance is unquantified. We measure juror independence and find that it does not hold, so the diversity constraint buys less than its statement implies. Our screener figures are high-variance. The jury-seed anchor removes grinding but not

timing, because a challenger who cannot choose the anchor can still choose when to file relative to checkpointing; fixing this needs a commit-then-reveal challenge flow rather than a patch. Our federation results are in-process and say nothing about network behaviour, clock skew or Byzantine validators. A multi-host testnet is the work that would change that. Juror availability falls as severity rises. Finally, our evaluation corpus is synthetic and small, and its labels were themselves a finding: eleven of thirty-two violation labels conflicted with the clause they were written from, caught only when models from nine vendors kept “failing” the same items. A sharper version of the remaining limit deserves stating on its own. The corpus items were written by language-model agents, so the items we label contested are contested for models at least; a corpus written and adjudicated by human authors, with a human baseline on the same 70 items, is the experiment we have not run, and it is the half of the shared-authorship confound still standing. The other half is closed: jurors from unrelated vendors reproduce the correlation on identical items at equal strength (§VII-C), so vendor lineage no longer discounts the magnitude and we no longer read our number as an upper bound. Prompt framing is the uncertainty our intervals do not cover — the original three-tier framing was unrecorded and its numbers are not strictly reproducible, which is why every verdict file now carries the hash of the prompt that produced it.

IX. RELATED WORK

Transparency logs and witness cosigning [1], [21] are the direct ancestors of our evidence plane and the verifiable-claims agenda [2] framed the problem. Training-side verification, including proof-of-learning [3] and compute accounting [4], complements our deployment-side scope, as do verifiable inference [5], recursive proof composition [29] and optimistic dispute games [6]. Our consensus assumptions rest on the PBFT lineage [7]–[9] and high-throughput DAG protocols [22], [23].

Oversight-by-AI draws on debate [10], constitutional methods [11] and LLM-as-judge evaluation [12]. PoLL [38] argues directly for the panel-of-judges design we adopt, finding that a panel of smaller models from disjoint families outperforms a single large judge at a fraction of the cost. Our result sits in tension with that conclusion without contradicting it, because PoLL compares a panel against an external large judge where we compare a jury against its own best constituent member. Both can hold. Concurrent work has since measured the underlying mechanism on generic benchmarks: correlated errors across 350+ models with explicit downstream analysis of LLM-as-judge evaluation [39] and a nine-judge panel across seven families carrying only about two independent votes of information, with the best single judge matching or beating the panel [40]. Our finding agrees with theirs and we take that as corroboration rather than as a claim of priority. Our delta is setting and response. We measure inside a protocol where a verdict slashes stake and moves registry status. We answer with an asymmetric voting rule and protected abstention. A complementary line fixes the aggregator instead, replac-

ing the mean with a robust estimator under contamination [41]. Decentralized dispute resolution pioneered the randomly drawn, staked, appealable jury for human jurors [31] and precedent-grounded decision support exists as retrieval tooling [32], where ours is a ledger object carrying verdict force. Hash-chained audit frameworks for LLM compliance [33], receiver-attested action receipts [34] and audit-trail tooling for accountability [42] supply base evidence layers. We build above them, in challenge, adjudication and consequence. Standards work is converging on the same substrate from the supply-chain and attestation directions, in transparent claim registration [43] and remote attestation architecture [44]. An adjacent governance line argues, as we do, for governing actions through institutional attestation rather than governing agents [45].

Blockchain-anchored federated learning [16], [17], [20] federates training under a shared ledger. We federate evidence about deployed conduct and deliberately avoid global consensus for it. Our components come from documentation and provenance practice [13]–[15], statistical auditing [24], sequential testing [25], differential privacy [26], threshold secret sharing [30] for break-glass escrow, assurance-contract mechanism design [27] for bootstrapping participation and output watermarking [28] as a complementary attribution channel where runtime instrumentation is absent. Policy instruments [18], [19], assurance-technology agendas [35], hardware-enabled guarantees [36] and treaty-scale verification proposals [37] define the deployment context we build for.

X. CONCLUSION

The deadlock in AI oversight is that everyone wants everyone else verified while the record stays with the audited. PACT makes deployed conduct non-repudiable and adjudicable at a cost that does not grow with traffic and keeps every individual action provable to a root no single party can rewrite. The mechanism most often proposed for scaling that oversight, panels of AI jurors, rests on an assumption that does not hold in the form usually stated, here and in concurrent work [39], [40]. Evaluator models do not fail independently. They fail together, on the cases where judgement matters. Majority voting over them can be worse than their best member. Neither finding kills the approach and both change what it must look like: asymmetric thresholds rather than symmetric ones, jurors selected for measured complementarity rather than declared diversity and humans retained as the only uncorrelated check in the system.

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